

Week 2 Homework

exercise

Sigma notation review: Write the sum in expanded form: a. $\sum_{i=1}^5 \sqrt{i}$ b. $\sum_{i=1}^n i^{10}$
c. $\sum_{j=0}^{n-1} (-1)^j$ d. $\sum_{j=n}^{n+3} j^2$

exercise

Sigma notation review: Write the sum in sigma notation: a. $1 + 2 + 3 + 4 + 5 + 6 + \dots + 10$
b. $2 + 4 + 6 + 8 + \dots + 2n$ c. $1 + x + x^2 + x^3 + \dots + x^n$ d. $1 + 2x + 3x^2 + 4x^3 + \dots + nx^{n-1}$

exercise

Factorial review: What is the factorial function? What notation is used for this function?

exercise

Factorial review: Calculate the following expressions using properties of the factorial function. a. $\frac{7!}{3!}$ b. $\frac{8!}{4!4!}$ c. $\frac{10!}{9!}$ d. $\frac{n!}{(n-2)!}$

exercise

Sequence Review: Write the first five terms of the following sequences: a. $a_1 = 1$, $a_{n+1} = \frac{1}{1+a_n}$ b. $b_1 = 0$, $b_2 = 1$, $b_n = b_{n-1} - b_{n-2}$ c. $c_1 = 1$, $c_n = c_{n-1} + n$ d. $d_1 = 1$, $d_n = 2d_{n-1}$

exercise

Sequence Review: Find a formula for the general term a_n of the given sequence:
a. $\{1, 4, 7, 10, 13, \dots\}$ b. $\{-1, 2, -6, 24, -120, \dots\}$ c. $\{1, 4, 9, 16, 25, \dots\}$ d. $\{0, 1, 3, 7, 15, \dots\}$